

DSD Final Checkpoint

Baseline checkpoint representation

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Issues Encountered

Reset input delay

The testbench changes reset at a different phase from the memory interface. Treating every input with the same delay caused misleading timing pressure and gate-level setup messages. The synthesis constraint was updated to use a separate reset input delay.

Cache data reset

Cache data arrays do not need to be reset because valid bits define whether a line can be used. Removing data-array reset reduced unnecessary reset mux paths while keeping valid/dirty/control reset for correctness.

Current status

The baseline processor supports the required checkpoint instruction set, handles structural/data/control hazards, includes cache flush support, and passes noHazard and hasHazard gate-level simulations at 3.00 ns.

Synthesis Result

Timing

Synthesis clock period	3.00 ns
Worst negative slack	0.00 ns
Total negative slack	0.00 ns
Violating paths	0
Logic levels	10

Gate simulation

Post-syn testbench cycle	3.00 ns
noHazard sim. time	1,407,896 ns
hasHazard sim. time	7,038,896 ns

Area summary

Combinational area	221,357.932724
Buf/Inv area	34,873.083013
Noncombinational area	178,868.619936
Macro/Black Box area	0.000000
Net interconnect area	3,749,825.752380
Total cell area	400,226.552660
Total area	4,150,052.305040

Gate-Level Simulation

```
[INFO]: Testing with pattern <noHazard>
-----
START!!! Simulation Start ....
-----
=====
Reset ...
=====
sh: ../00_TESTBED/info/success: Permission denied
=====

\(^o^)/ CONGRATULATIONS!! The simulation result is PASS!!

=====
$finish called from file "../00_TESTBED/testbench/Final_tb.v", line 247.
$finish at simulation time          1407896
```

noHazard pattern

```
[INFO]: Testing with pattern <hasHazard>
-----
START!!! Simulation Start ....
-----
=====
Reset ...
=====
sh: ../00_TESTBED/info/success: Permission denied
=====

\(^o^)/ CONGRATULATIONS!! The simulation result is PASS!!

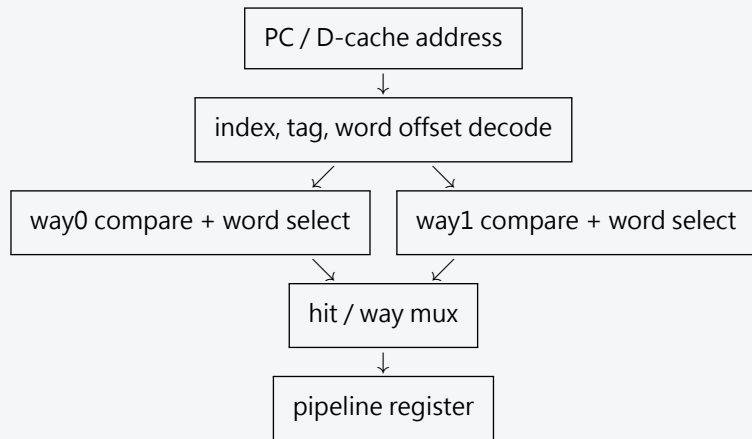
=====
$finish called from file "../00_TESTBED/testbench/Final_tb.v", line 247.
$finish at simulation time          7038896
```

hasHazard pattern

Both baseline gate-level simulations pass with SDF annotation at 3.00 ns.

Critical Path and Future Plan

Cache hit datapath



Observation

Cache access and branch redirection paths are important timing targets for final performance.

Next steps

- Experiment with cache block count and way count.
- Tune I-cache/D-cache size and associativity.
- Experiment with branch prediction algorithms.
- Explore compressed and multiplication extensions.